

MambaRefine-CD

Mamba Vision with Region-Boundary Temporal Refinement

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June 1, 2026

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Introduction & Motivation

The Challenge of Binary Change Detection

Binary Change Detection (CD) in remote sensing aims to identify changed regions between bi-temporal images.

Core Challenges:

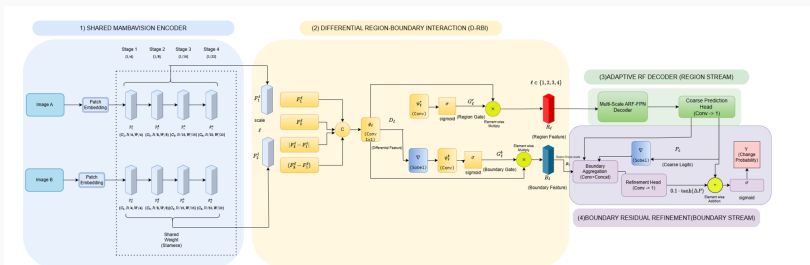
- **Nuisance Variation:** Must be invariant to illumination shifts, seasonal changes, and shadows.
- **Boundary Degradation:** Errors are most visible near object boundaries, resulting in shifted, fragmented, or over-smoothed masks.
- **Temporal Polarity Loss:** Standard absolute differencing captures magnitude but loses direction (appearance vs. disappearance).

Our architecture frames the problem as a three-question system:

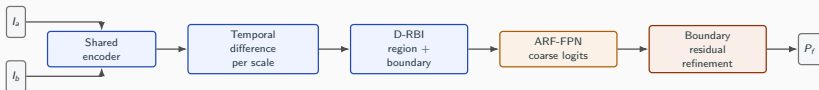
1. **What changed?** Handled by constructing rich temporal feature differences.
2. **Where is the changed region?** Handled by the Region Stream and an Adaptive Receptive-Field (ARF) Decoder.
3. **Where exactly is the boundary?** Handled by the Boundary Stream and bounded residual refinement.

Architecture Overview

MambaRefine-CD Architecture

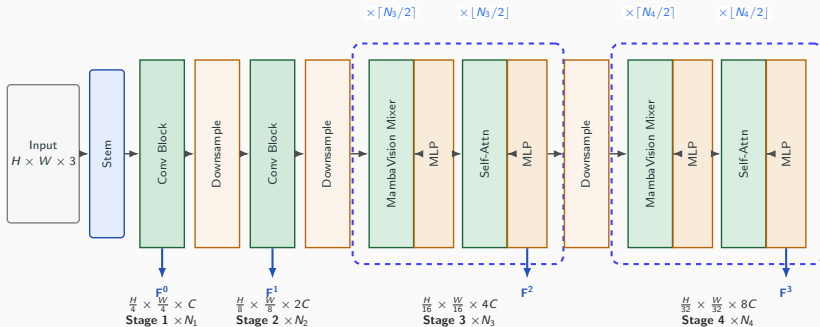


High-Level Data Flow



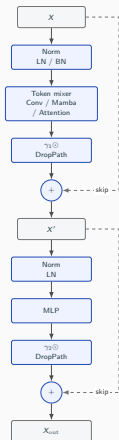
$$\begin{aligned}\{F_a^s\}_{s=0}^3 &= E_\theta(I_a), & \{F_b^s\}_{s=0}^3 &= E_\theta(I_b), \\ \Delta^s &= F_b^s - F_a^s, & D_{\text{in}}^s &= [|\Delta^s|; \Delta^s] \text{ for abs_signed}, \\ (R^s, B^s) &= \text{DRBI}_s(D_{\text{in}}^s), & P_c &= \text{ARFFPN}(\{R^s\}_{s=0}^3), \\ P_f &= P_c + 0.1 \tanh(\Delta P_{\text{raw}}).\end{aligned}$$

Shared MambaVision Encoder: Four Stages



Encoder outputs are tapped after each stage: $F^s \in \mathbb{R}^{B \times C2^s \times H/2^{s+2} \times W/2^{s+2}}$. For MambaVision-S:
 $C = 96$, $N = (3, 3, 7, 5)$.

Encoder Block Equations



$$U = \mathcal{M}(\text{Norm}_1(X)),$$

$$X' = X + \text{DropPath}(\gamma_1 \odot U),$$

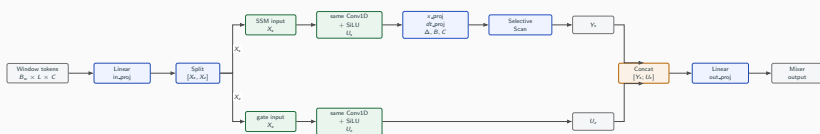
$$V = \text{MLP}(\text{Norm}_2(X')),$$

$$X_{out} = X' + \text{DropPath}(\gamma_2 \odot V).$$

$$\mathcal{M} = \begin{cases} \text{convolutional residual stack,} & \text{stages 1-2,} \\ \text{MambaVision mixer or self-attention,} & \text{stages 3-4.} \end{cases}$$

The block is pre-normalized and residual at both the token-mixing and MLP sublayers.

MambaVision Mixer Inside Stages 2–3



$$[X_s, X_z] = \text{split}(W_{\text{in}}X),$$

$$U_z = \text{SiLU}(\text{Conv1D}_{\text{same}}(X_z)),$$

$$h_k = \bar{A}_k h_{k-1} + \bar{B}_k U_{s,k},$$

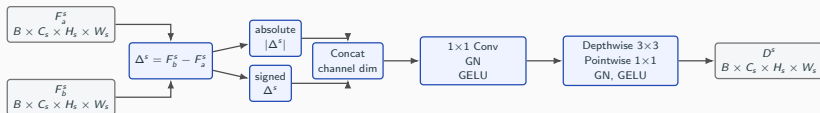
$$U_s = \text{SiLU}(\text{Conv1D}_{\text{same}}(X_s)),$$

$$[\Delta_k, B_k, C_k] = W_x U_{s,k},$$

$$Y = W_{\text{out}}[Y_s; U_z].$$

Differential Region-Boundary Interaction (D-RBI)

D-RBI 2.1: Difference Input and 1x1 Projection

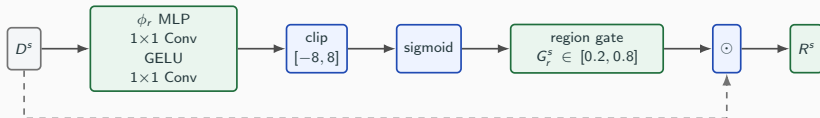


$$\Delta^s = F_b^s - F_a^s, \quad D_{\text{in}}^s = [|\Delta^s|; \Delta^s] \quad \text{for default abs.signed,}$$

$$\hat{D}^s = \text{GELU}(\text{GN}(W_p^{1 \times 1} * D_{\text{in}}^s)),$$

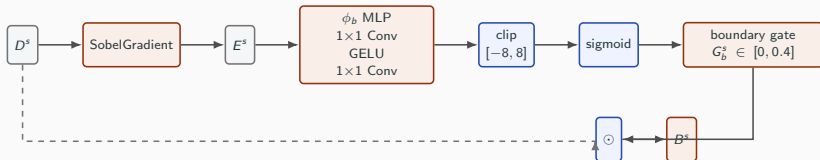
$$D^s = \text{GELU}(\text{GN}(W_{\text{pw}}^{1 \times 1} * (W_{\text{dw}}^{3 \times 3} * \hat{D}^s))).$$

D-RBI 2.2: Regional Feature Path



$$\begin{aligned} H_r^s &= \text{GELU}(W_{r,1}^{1 \times 1} * D^s + b_{r,1}), \\ \phi_r(D^s) &= W_{r,2}^{1 \times 1} * H_r^s + b_{r,2}, \\ G_r^s &= 0.2 + 0.6 \sigma(\text{clip}(\phi_r(D^s), -8, 8)), \quad R^s = D^s \odot G_r^s. \end{aligned}$$

D-RBI 2.3: Boundary Feature Path



$$\begin{aligned} E^s &= \text{SobelGradient}(D^s), & H_b^s &= \text{GELU}(W_{b,1}^{1 \times 1} * E^s + b_{b,1}), \\ \phi_b(E^s) &= W_{b,2}^{1 \times 1} * H_b^s + b_{b,2}, \\ G_b^s &= 0.4 \sigma(\text{clip}(\phi_b(E^s), -8, 8)), & B^s &= D^s \odot G_b^s. \end{aligned}$$

D-RBI 2.4: Sobel Block

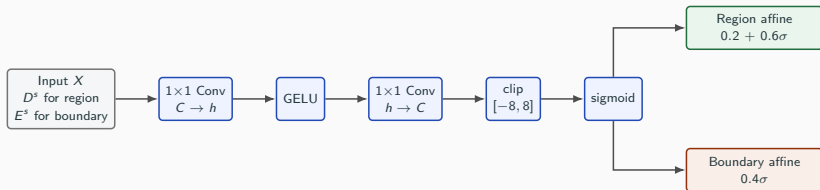


$$K_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}, \quad K_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

$$G_x = x * K_x, \quad G_y = x * K_y$$

$$\text{SobelGradient}(x) = \text{clip} \left(\sqrt{G_x^2 + G_y^2 + 10^{-6}}, 0, 10 \right).$$

D-RBI 2.5: ϕ_r and ϕ_b Gate Blocks

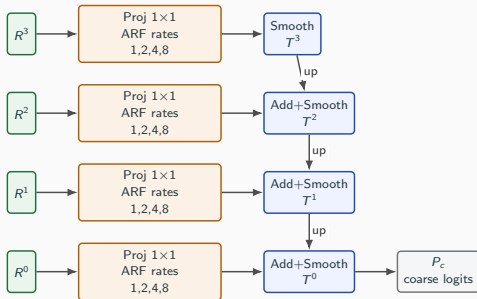


$$h = \max(\lfloor 0.25C \rfloor, 1), \quad \phi(X) = W_2^{1 \times 1} * \text{GELU}(W_1^{1 \times 1} * X + b_1) + b_2.$$

$$G_r = 0.2 + 0.6\sigma(\text{clip}(\phi_r, -8, 8)), \quad G_b = 0.4\sigma(\text{clip}(\phi_b, -8, 8)).$$

Decoding & Refinement

Multi-Scale ARF-FPN Decoder



$$\tilde{R}^s = \text{Proj}_s^{1 \times 1}(R^s),$$

$$Z_{s,r} = \text{ConvNormGELU}_{3 \times 3, d=r}(\tilde{R}^s),$$

$$\alpha_s = \text{softmax}(\text{MLP}(\text{GAP}(\tilde{R}^s))),$$

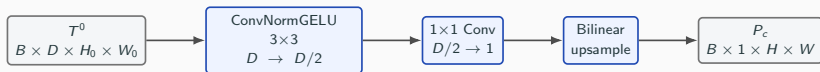
$$P^s = \sum_{r \in \{1,2,4,8\}} \alpha_{s,r} Z_{s,r},$$

$$T^3 = \text{Smooth}_3(P^3),$$

$$T^s = \text{Smooth}_s(P^s + \text{Up}(T^{s+1})),$$

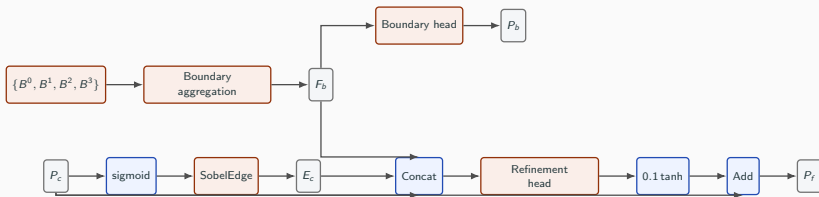
$$P_c = \text{Up}(\text{Head}(T^0)).$$

Coarse Prediction Head



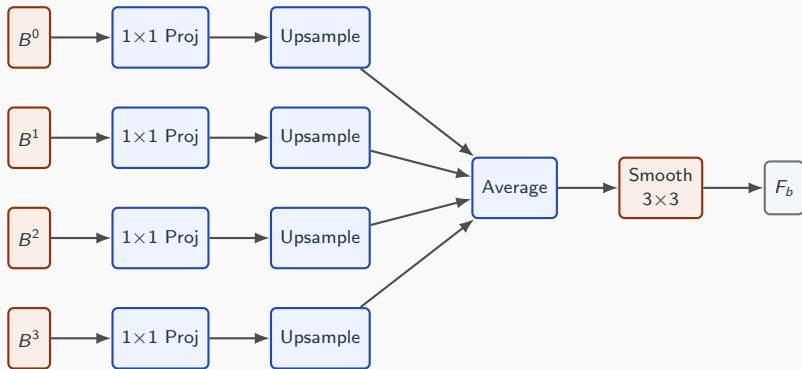
$$H_c = \text{GELU}(\text{GN}(W_h^{3 \times 3} * T^0)), \quad \ell_c = W_o^{1 \times 1} * H_c + b_o, \quad P_c = \text{Up}(\ell_c; H, W).$$

Boundary Residual Refinement: Full Block



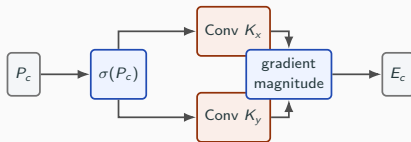
$$P_b = \text{BoundaryHead}(F_b), \quad E_c = \text{SobelEdge}(\sigma(P_c)), \quad P_f = P_c + 0.1 \tanh(\text{RefineHead}([F_b; P_c; E_c])).$$

5.1 Boundary Aggregation Block



$$Q^s = \text{Up}(\text{Proj}_{b,s}^{1 \times 1}(B^s); H, W), \quad F_b = \text{Smooth} \left(\frac{1}{4} \sum_{s=0}^3 Q^s \right).$$

5.2 Sobel Operator for Coarse Probability

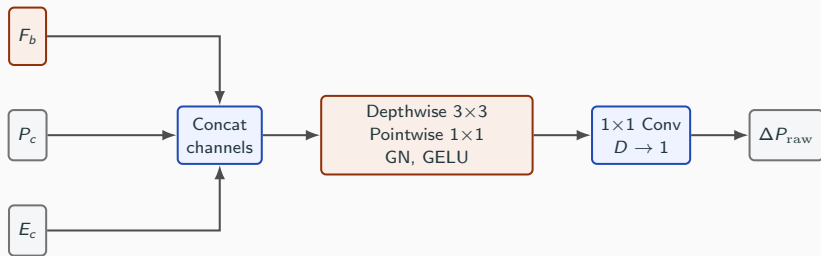


$$P = \sigma(P_c), \quad G_x = P * K_x, \quad G_y = P * K_y$$

$$E_c = \sqrt{G_x^2 + G_y^2 + 10^{-6}}.$$

This Sobel block is single-channel and is applied to the coarse change probability, not to the RGB image.

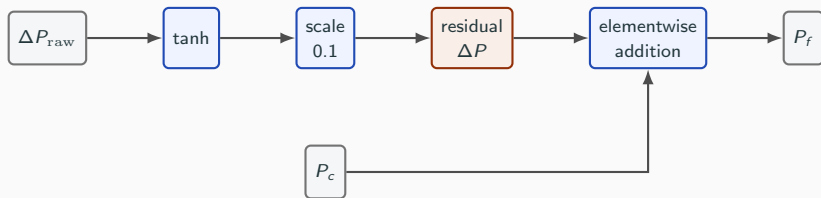
5.3 Refinement Head



$$X_r = [F_b; P_c; E_c] \in \mathbb{R}^{B \times (D+2) \times H \times W},$$

$$H_r = \text{GELU}(\text{GN}(W_{\text{pw}}^{1 \times 1} * (W_{\text{dw}}^{3 \times 3} * X_r))), \quad \Delta P_{\text{raw}} = W_{\Delta}^{1 \times 1} * H_r + b_{\Delta}.$$

5.4 Elementwise Addition Block



$$\Delta P = 0.1 \tanh(\Delta P_{\text{raw}}), \quad P_f = P_c + \Delta P.$$

The residual is bounded in logit space, so refinement can sharpen boundaries without rewriting the coarse region prediction.

The model is supervised at multiple stages:

$$\mathcal{L} = \mathcal{L}_{\text{BCE}} + \mathcal{L}_{\text{Dice}} + 0.4\mathcal{L}_{\text{coarse}} + 0.1\mathcal{L}_{\text{boundary}}$$

- **Main Loss:** BCE and Dice supervise the final prediction z_f .
- **Coarse Loss:** Stabilizes the main region decoder.
- **Boundary Loss:** Uses a Sobel-edge l_1 loss to force contour consistency.

Main Results

Main Test Results

Dataset	Model	Pre.	Rec.	F1	IoU	OA
DSIFN-CD	MambaRefine-CD	95.47	95.87	95.67	91.71	96.98
WHU-CD	MambaRefine-CD	95.79	94.90	95.34	91.10	99.56
WHU-CD	MambaRefine-CD VMamba	96.32	94.68	95.49	91.37	99.57

What this says

- DSIFN reaches 95.67 F1 and 91.71 IoU on the clean split.
- WHU reaches 95.34 F1 / 91.10 IoU with the MambaRefine-CD result summary.
- The VMamba variant gives the best WHU score: 95.49 F1 / 91.37 IoU.

Evaluation rule

- Test metrics use the EMA/best checkpoint selected by validation F1.
- The threshold is selected from $\{0.30, 0.35, \dots, 0.60\}$.
- DSIFN split integrity was verified as PASS; no train/test overlap.

Dataset and Training Protocol

Data

- DSIFN-CD: 2758 train, 394 val, 789 test images.
- DSIFN test: 3156 tiles at 256×256 , overlap 0.25.
- WHU-CD: standard train/val/test split.

Inference

- Patch-based inference with crop 256.
- Validation every 5000 iterations.
- Final mask: $\hat{Y} = \mathbb{K}[\sigma(P_f) > \tau^*]$.

Setting	Value
Optimizer	AdamW
Learning rate	5×10^{-5}
LR schedule	cosine decay
Warmup	2500 iters
Weight decay	0.01
Grad clip	0.5
Batch / image	8 / 256×256
Precision	AMP BF16/FP16
EMA decay	0.999
Augmentation	horizontal + vertical flip

Metric Equations

Let changed pixels be the positive class after thresholding $\hat{Y} = \mathbb{K}[\sigma(P_f) > \tau]$.

With $\epsilon = 10^{-6}$:

$$\text{Precision} = \frac{TP}{TP + FP + \epsilon} \times 100,$$

$$\text{Recall} = \frac{TP}{TP + FN + \epsilon} \times 100,$$

$$F1 = \frac{2 \text{ Precision Recall}}{\text{Precision} + \text{Recall} + \epsilon},$$

$$\text{IoU} = \frac{TP}{TP + FP + FN + \epsilon} \times 100,$$

$$\text{OA} = \frac{TP + TN}{TP + TN + FP + FN + \epsilon} \times 100.$$

Boundary metrics

$$\text{BF1} = \frac{2P_b R_b}{P_b + R_b + \epsilon}, \quad \text{BIOU} = \frac{|B_{\text{pred}} \cap B_{\text{gt}}|}{|B_{\text{pred}} \cup B_{\text{gt}}| + \epsilon} \times 100.$$

BF1, *BIOU*, and Trimap F1 are computed on edge/trimap regions, so they measure contour alignment rather than only changed-area overlap.

SOTA Model Results

SOTA Results on DSIFN-CD

Model	Pre.	Rec.	F1	IoU	OA
FC-Siam-Conc	66.45	54.21	59.71	42.56	87.57
SNUNet	60.60	72.89	66.18	49.45	87.34
ChangeFormer	88.48	84.94	86.67	76.48	95.56
BiFA	73.99	68.87	71.34	55.45	90.80
FTAN	90.54	88.61	89.56	81.10	–
ADSFNet	94.79	95.24	95.01	90.50	98.30
Mamba-CD	95.60	95.61	95.61	91.69	98.51
MambaRefine-CD	95.47	95.87	95.67	91.71	96.98

- MambaRefine-CD gives the highest DSIFN F1 and IoU in the provided result summary.
- The gain over Mamba-CD is small in changed-region metrics, so boundary quality and ablation evidence are important for the method claim.

SOTA Results on WHU-CD

Method	Pre.	Rec.	F1	IoU	OA
FC-EF	71.63	67.25	69.37	53.11	97.61
STANet	79.37	85.50	82.32	69.95	98.52
SNUNet	85.60	81.49	83.50	71.67	98.71
IFNet	96.91	73.19	83.40	71.52	98.83
ChangeFormer	91.83	88.02	89.88	81.63	99.12
BiFA	95.15	93.60	94.37	89.34	99.56
RSM-CD	93.37	90.42	91.87	84.96	—
SChanger	94.62	91.83	93.20	87.27	—
CDMamba	95.58	92.01	93.76	88.26	99.51
Mamba-CD	96.52	93.91	95.20	90.83	99.62
MambaRefine-CD	95.79	94.90	95.34	91.10	99.56
MambaRefine-CD VMamba	96.32	94.68	95.49	91.37	99.57

The VMamba variant is the strongest WHU result in this summary; the MambaVision result has the highest recall among the top group.

Changed-region accuracy

- DSIFN: ours improves over Mamba-CD by +0.06 F1 and +0.02 IoU.
- WHU: MambaRefine-CD improves over Mamba-CD by +0.14 F1 and +0.27 IoU.
- WHU VMamba improves over Mamba-CD by +0.29 F1 and +0.54 IoU.

Why still useful

- D-RBI explicitly separates region and boundary evidence.
- ARF-FPN adapts receptive field per scale instead of using one fixed context size.
- Boundary residual correction is bounded in logit space, so it sharpens edges without rewriting the whole mask.

Ablation Study Results

Component Ablation on DSIFN-CD

ID	Variant	Main added component	Params (M)	F1	IoU	OA
A0	FPN Baseline	SimpleCNN encoder	7.84	76.63	62.12	83.80
A1	MambaVision-S + FPN	MambaVision encoder	53.54	93.21	87.28	95.72
A2	+ D-RBI unsigned	Absolute-diff D-RBI	54.98	93.33	87.50	95.35
A3	+ Signed Diff	Signed temporal stream	55.34	94.28	89.19	95.99
A4	+ ARF-FPN	Adaptive RF decoder	65.12	94.36	89.32	96.05
A5	+ Boundary Residual	Bounded residual head	65.19	93.59	87.94	95.53

This table isolates architectural additions before the final full-loss configuration. The final DSIFN result is reported separately as 95.67 F1 / 91.71 IoU.

Ablation Deltas: What Actually Moved the Score

Change	$\Delta F1$	ΔIoU	
A0 $\rightarrow$ A1: MambaVision backbone	+16.58	+25.16	The backbone is the dominant region-accuracy gain.
A1 $\rightarrow$ A2: unsigned D-RBI	+0.12	+0.22	
A2 $\rightarrow$ A3: signed difference	+0.95	+1.69	D-RBI gives a small unsigned gain; signed temporal evidence gives the larger D-RBI gain.
A3 $\rightarrow$ A4: ARF-FPN	+0.08	+0.13	
A4 $\rightarrow$ A5: BRR only	-0.77	-1.38	

- ARF-FPN gives a small region-score gain but improves boundary metrics later.
- Boundary residual refinement is harmful when it is added without the final supervision schedule.

Boundary-Aware Ablation

Variant	F1	IoU	BF1	BloU	Trimap F1 3px
A1 MambaVision-FPN	93.21	87.28	61.36	38.41	68.02
A4 + ARF-FPN	93.71	88.16	65.94	42.06	70.27
A5 + Boundary residual	93.59	87.94	65.14	41.37	69.98
A6 Full model	94.58	89.72	71.94	47.39	72.93
WHU Full model	95.15	90.76	89.49	71.37	87.01

A1 → A6 improves BF1 by +10.58 and BloU by +8.98 on the boundary-eval set. A5 shows the key bottleneck: adding a boundary residual head without boundary supervision does not guarantee better contours.

Backbone Scaling and Efficiency

Backbone	Params (M)	GFLOPs	F1	IoU	OA
MambaVision-T	46.73	50.05	94.58	89.72	96.24
MambaVision-S	65.40	64.08	95.67	91.71	96.98
MambaVision-B	113.37	109.15	95.77	91.89	97.07
MambaVision-L	244.76	222.95	—	—	—

Small $\rightarrow$ Base gives only $+0.10$ F1 for $1.73\times$ parameters. MambaVision-S is therefore the best accuracy/complexity point in the completed runs.

GFLOPs are an fvc core relative profile; selective scan and attention operators were reported as unsupported, so the profile is best used for relative scaling, not exact hardware throughput.

Temporal Input Ablation: WHU-CD VMamba-Tiny

ID	Temporal input	Pre.	Rec.	F1	IoU	OA
T0	features only	96.44	94.43	95.43	91.25	99.56
T1	signed only	96.43	94.24	95.32	91.07	99.55
T2	abs only	95.99	94.67	95.32	91.06	99.55
T3	features signed	96.14	94.62	95.37	91.15	99.56
T4	features abs	96.99	94.78	95.87	92.07	99.61
T5	signed abs	96.02	94.75	95.38	91.17	99.56
T6	full	95.59	95.29	95.44	91.27	99.56

On WHU with VMamba-Tiny, the best temporal construction is *features + abs*. The full input has the highest recall but not the best F1/IoU.

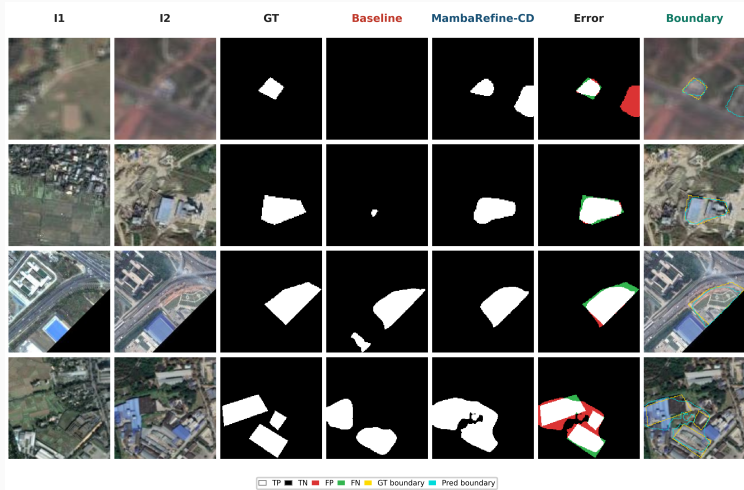
Temporal Difference Ablation: DSIFN-CD Group A

Mode	F1	IoU	OA
Absolute only	94.655	89.853	96.291
Signed only	94.718	89.966	96.322
Absolute + signed	94.587	89.731	96.237

- Signed-only is +0.13 F1 and +0.24 IoU above abs+signed in this DSIFN Group-A test.
- The modes are close, so temporal mode alone is not the dominant bottleneck.
- The model is more sensitive to how temporal evidence is gated, decoded, and supervised.

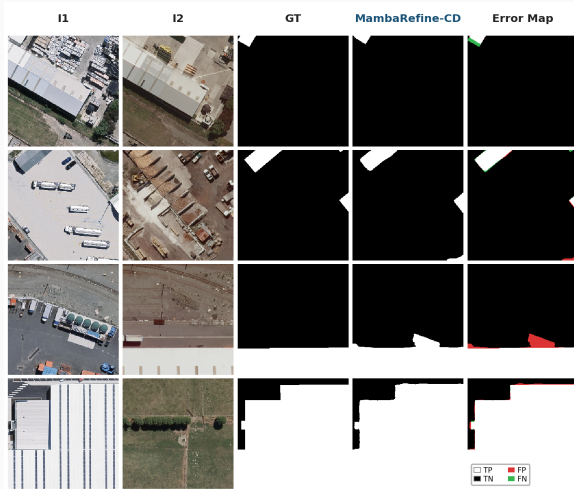
Qualitative Results

Qualitative Results: DSIFN-CD



The qualitative figure is used to verify whether the metric gains correspond to cleaner building boundaries and fewer fragmented missed-change regions.

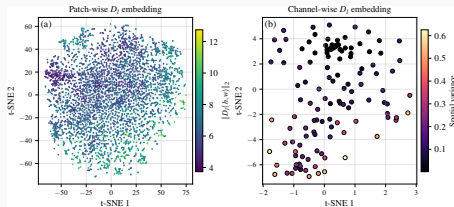
Qualitative Results: WHU-CD



WHU examples mainly stress building-boundary precision because overall accuracy is already high from the large unchanged background.

Bottleneck Diagnostics

Bottleneck Diagnostic: 1x1 Compression

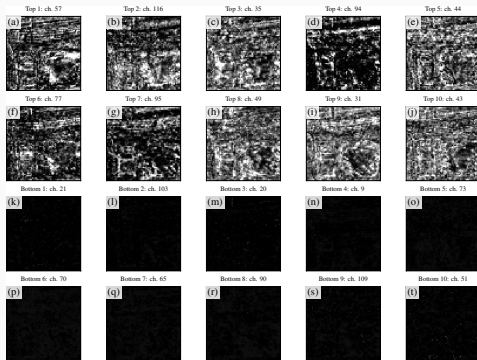


$$\rho = \frac{\|D_I\|_2}{\|X_\Delta\|_2}, \quad c = \cos(WX_\Delta, D_I)$$

Model	ρ	c
Vanilla	3.059	0.483
Absolute-only	0.602	0.771
Signed-only	0.614	0.756

The vanilla bottleneck is not a passive channel reducer; it strongly re-encodes temporal evidence. This explains why downstream gates and losses matter.

Bottleneck Diagnostic: Channel Activation Hierarchy



Spatial variance per channel:

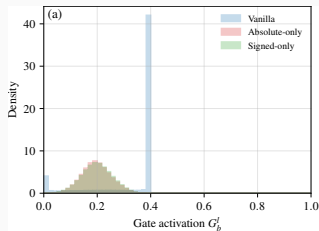
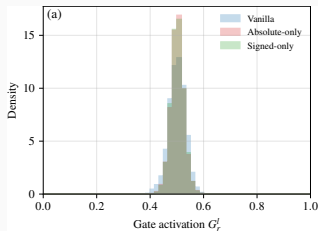
$$v_c = \frac{1}{HW} \sum_{i,j} (D_{I,cij} - \mu_c)^2.$$

Channel group	Variance range
Top-10 active	0.435–0.626
Bottom-10 quiet	0.0015–0.0028

High-variance channels localize structured change evidence.

Low-variance channels behave like background suppression or weak context carriers.

Bottleneck Diagnostic: Region and Boundary Gates



Model / gate	min	max	mean	$G < 0.05$
Vanilla G_r	0.364	0.629	0.501	0.000
Vanilla G_b	0.000	0.400	0.329	0.086
Abs-only G_b	0.013	0.389	0.197	0.001
Signed-only G_b	0.008	0.391	0.199	0.003

Region gates stay centered near 0.5 and do not saturate. Boundary gates are intentionally capped at 0.4, so the boundary path refines rather than dominates the region path.

Experimental Bottlenecks Found in the Archive

- **Boundary residual without boundary supervision:** A4 $\rightarrow$ A5 decreases F1 and IoU. The residual branch needs edge loss and coarse supervision, not just the main mask loss.
- **Unsigned D-RBI is insufficient:** A1 $\rightarrow$ A2 drops in the averaged table. Signed temporal evidence recovers and improves the D-RBI path.
- **Backbone scaling saturates:** Small $\rightarrow$ Base adds $1.73\times$ parameters for only $+0.10$ F1.
- **Direct SOTA reproduction was incomplete:** archived jobs hit missing/manual checkpoints, DSIFN dataset-layout mismatch for ChangeFormer, and a WHU CUDA OOM case.
- **Profiling is relative:** fvcare reported unsupported selective-scan and attention operators, so FLOPs are useful for variant scaling but not exact runtime accounting.
- **Legacy result hygiene matters:** wrong-backbone and different-evaluator runs were excluded from canonical averages.

Conclusion

- **Main result:** DSIFN-CD reaches 95.67 F1 / 91.71 IoU; WHU-CD reaches 95.34 F1 / 91.10 IoU, with the VMamba variant at 95.49 F1 / 91.37 IoU.
- **Mechanism:** MambaVision gives the major region gain; signed D-RBI and ARF-FPN provide smaller but measurable additions.
- **Boundary value:** A1 $\rightarrow$ A6 improves BF1 by 10.58 points and BloU by 8.98 points in boundary evaluation.
- **Practical choice:** MambaVision-S is the best completed accuracy/complexity trade-off; larger backbones show diminishing returns.

Questions?

Thank you for your time.